YAKEEN

NEET 2.0 2027

Physics | Lecture - 7

Motion in a Plane

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Todays Goal

Realive motion in 1-D

H/w goal

1. S.A → 1 & 2
   2-D
2. Mahamanthan 1-D ✓
3. Revision of 2-D motion through Work Book from Lectur (1) to (6)

→ rough me Likh Ke revise

Revision and Work-Book

of 2-D

Equation of Trajectory → Relation between x- and y - Co-ordinate of motion

MR* Box
$x(t)$, $y(t)$, $v(t)$, $a(t)$, Kuch bhi given ho, time ko hata ke x & y ka relation nikalo.

MR* Box → Graph b/w x and y represent Path, but v/t, x/t & a/t graph does not represent Path of motion.

MR* Box
equation → Path
$y = mx + c$ → str.
$y = x^2$ →
$x^2 + y^2 = R^2$ →
$x^2/a^2 + y^2/b^2 = 1$ →

① $x = 8\sin(t)$
$y = 4\sin(t)$

② $x = 3t$
$y = 9t$

③ $y = 3t^2$
$x = 2t$

④ $x = 3\sin(t)$
$y = 3\cos(t)$

⑤ $x = 4\sin(t)$
$y = 5\cos(t)$

⑥ $\vec{r} = 2t\,\hat{i} + 4t^2\,\hat{j}$

⑦ $V = 2i + x\,\hat{j}$

General 2-D Motion

$$\vec{r} = x\hat{i} + y\hat{j}$$

Diff^{n} w.r.t. time

$$\frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$$

$$\boxed{\vec{v} = v_x\hat{i} + v_y\hat{j}}$$

Magnitude of velocity

speed $v = \sqrt{v_x^2 + v_y^2}$

Direction of motion.

$$\tan\theta = \frac{v_y}{v_x}$$

$$\vec{v} = v_x\hat{i} + v_y\hat{j}$$

Diff^{n} w.r.t. time

$$\frac{d\vec{v}}{dt} = \frac{d\vec{v}_x}{dt}\hat{i} + \frac{d\vec{v}_y}{dt}\hat{j}$$

$$\vec{a} = a_x\hat{i} + a_y\hat{j}$$

Motion in plane with Constant acc^{n}. { Equation of motion }

x-axis	y-axis
$\vec{V}_x = \vec{u}_x + \vec{a}_x t$	$\vec{V}_y = \vec{u}_y + \vec{a}_y t$
$\vec{S}_x = \vec{u}_x + \vec{a}_x t$	$\vec{S} = \vec{u} + \vec{a} t$
$V_x^2 - u_x^2 = 2a_x S_x$	$v^2 - u^2 = 2as$

MR* Box

→ question se sare physical quantity ko x and Y axis me seprate karo (component)

→ x and y me seprate equation of motion lagao

→ Vector ke concept se vmko add karo

mR* Box

$\vec{a}_x = 0$ $\vec{V}_x = \text{Cost}^n$

uniform motion in x axis

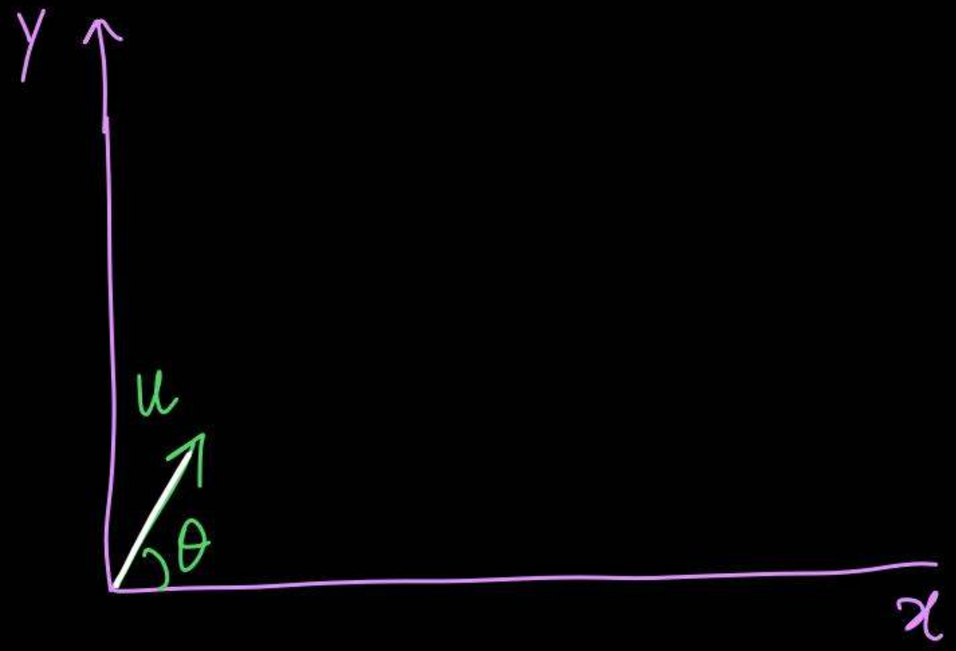
$\vec{a}_y \neq 0$ $\vec{V}_y = \text{varible}$

Non-uniform Motion

(Q) Position $\vec{r} = 3t^2\hat{i} + 2t\hat{j}$ find velocity and acceleration at $t = 2$ sec

(Q) velocity of object $\vec{v} = 3\hat{i} + 4\hat{j}$ and $\vec{a} = 0.4\hat{i} + 0.3\hat{j}$ then find velocity after $t = 10$ sec

Projectile motion : 2D Non uniform motion [variable velocity] with uniform acceleration.



Physical Quantity	x-axis	y-axis	Net
			–
initial u_x =			
acc^n			
$\vec{V}_t$			
disp^m.			
eqn. of th.			

① Velocity of maximum height :- $u_x = u\cos\theta$.

② Velocity of collision :-

③ change in velocity in compt motion :-

④ Average velocity in compt motion

⑤ velocity at time 't'

⑥ Angle of Projection

⑦ Direction of motion at time 't'

⑧ Time of flight

⑨ maximum Height

(10) Horizontal Range

(11) Time when velocity is ⊥r to intial velocity of Projection

(12) equation of Trajectory

(13) Relation between H and R

(14) Condition of maximum Range :-

(15) if air Resistance in x-axis only

(16) Angle B/w acc^n & velocity at maximum Height

(17) Tangential and centripetal acc^n at maximum Height

(18) complementry Angle :-

(19) Ball is projected with same speed at α and β such that Range are same then :-

$$\frac{H_1}{H_2} =$$

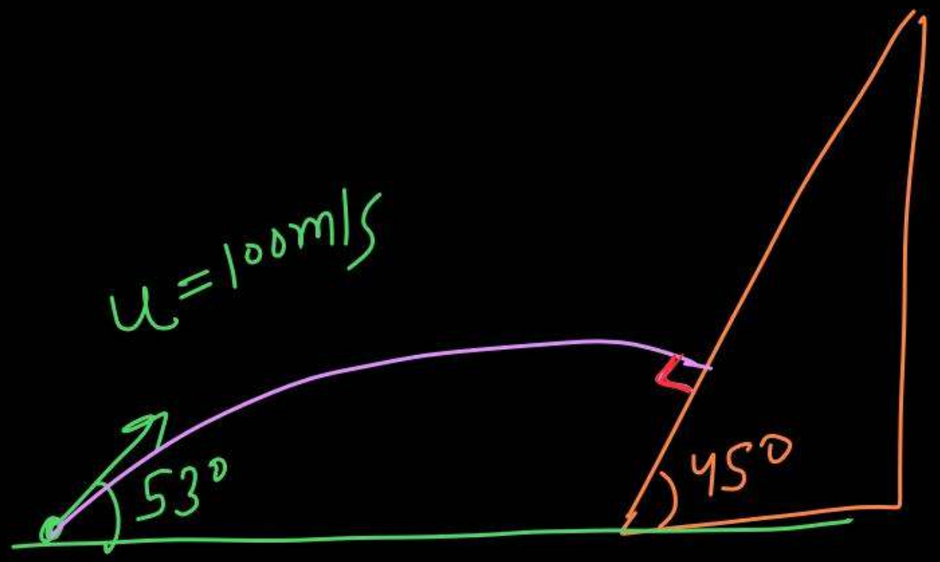
$$\frac{T_1}{T_2} =$$

(20) Radius of curvature at maximum Height :-

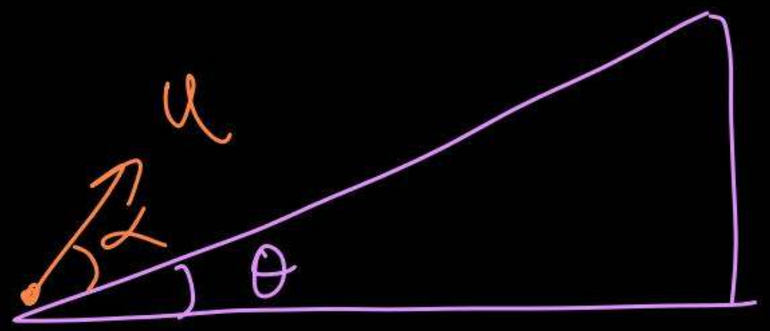
Object is Projected at $\vec{u} = 30\hat{i} + 40\hat{j}$ then find :-

Sab Kuchh hh ...

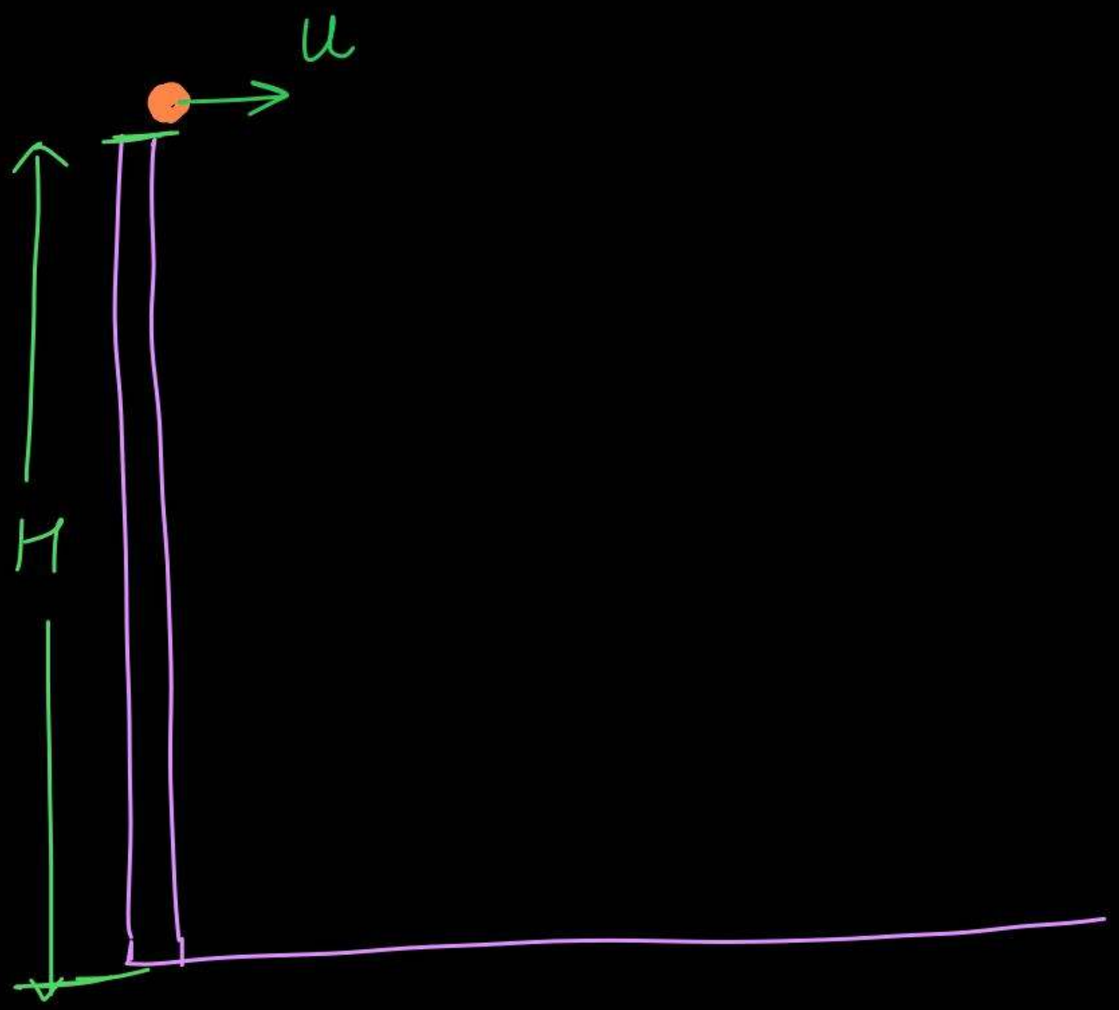
Time of collsion ??



Projectile on Inclined:-



Horizontal Projectile Motion



x-axis

$\vec{u_x} = u$

$a_x = 0$

$\vec{v_x} = u$

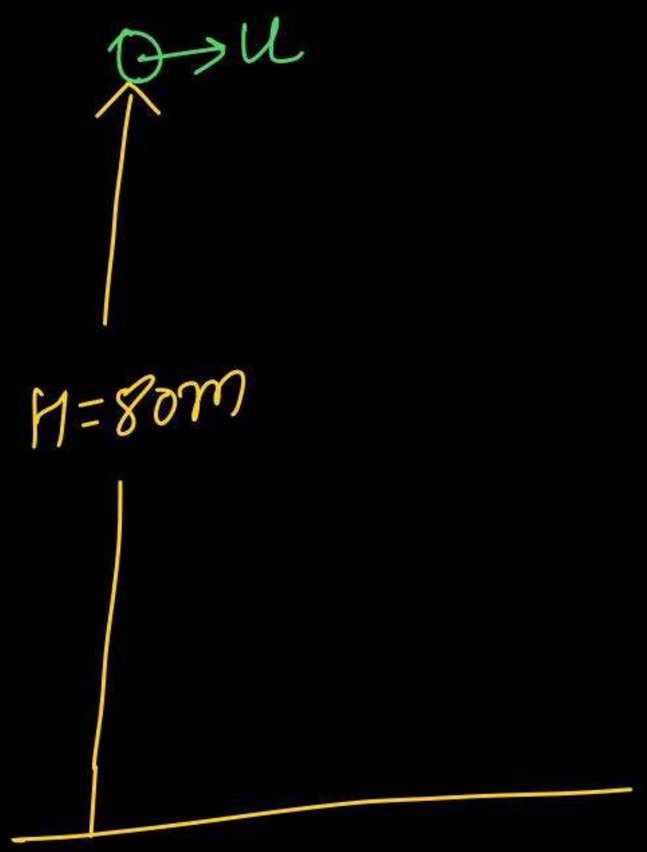
uniform motion in x

Y-axis

$u_y = 0$

$a_y = g$

$v_y = gt\,\hat{j}$



(1) Time of flight

(2) Range

(3) Maximum Height

(4) velocity at t=4sec

(5) dirn of motion at t=4sec

(6) Position of object at $t = 3\,sec$

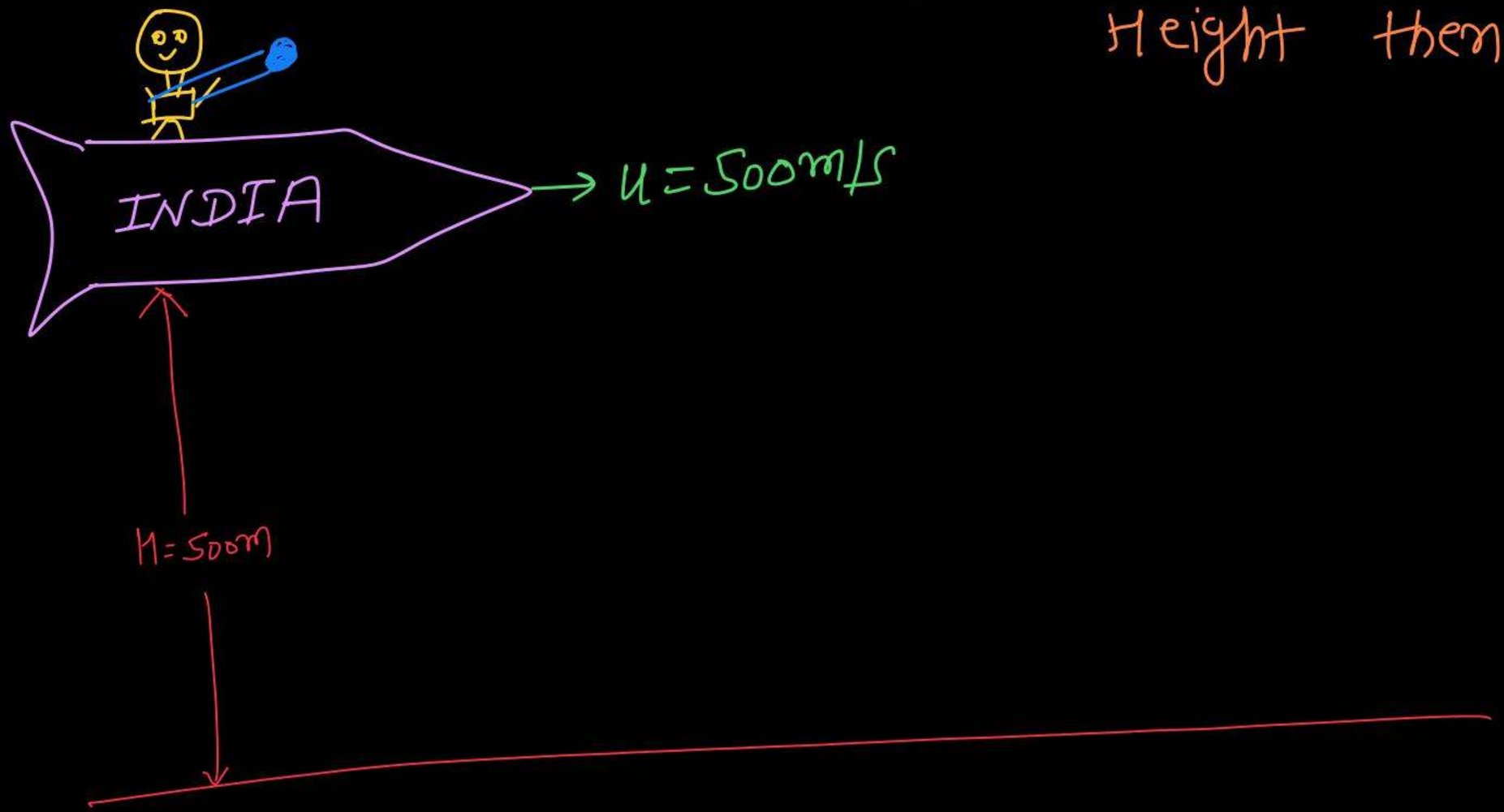
(7) Position vector at t

(8) Equation of Trajectory

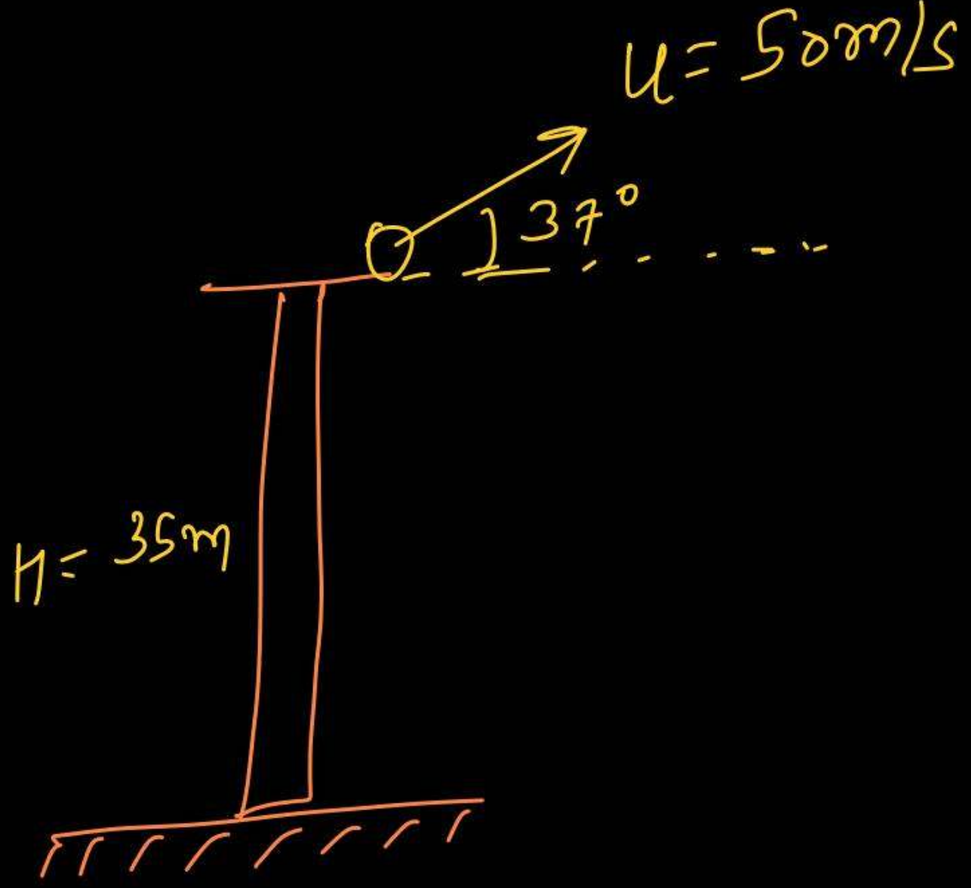
(9) velocity at ground

(10) Angle of collsion at ground from Horizontal :-

bomb is droped from 500m Height then Range of bomb



(Q) find Time of flight and Range :-



Todays

class

Realive motion

* $\vec{v}_{AB} = \vec{v}_{AG} - \vec{v}_{BG}$

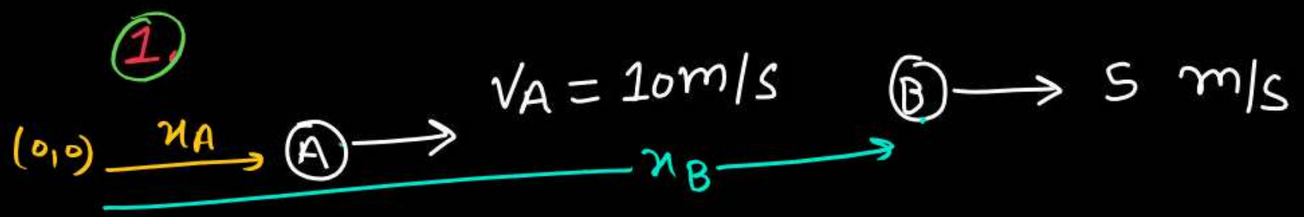
* $\vec{v}_{AB} = \vec{v}_{A\,lift} - \vec{v}_{B\,lift}$

$\vec{V}_{AB} = (\vec{v}_{AG} - \vec{v}_{lift\,G}) - (\vec{v}_{BG} - \vec{v}_{lift\,G})$

* $\vec{a}_{AB} = \vec{a}_{AG} - \vec{a}_{BG}$

$\vec{v}_{AB} = \vec{v}_{AG} - \vec{v}_{BG}$

1.



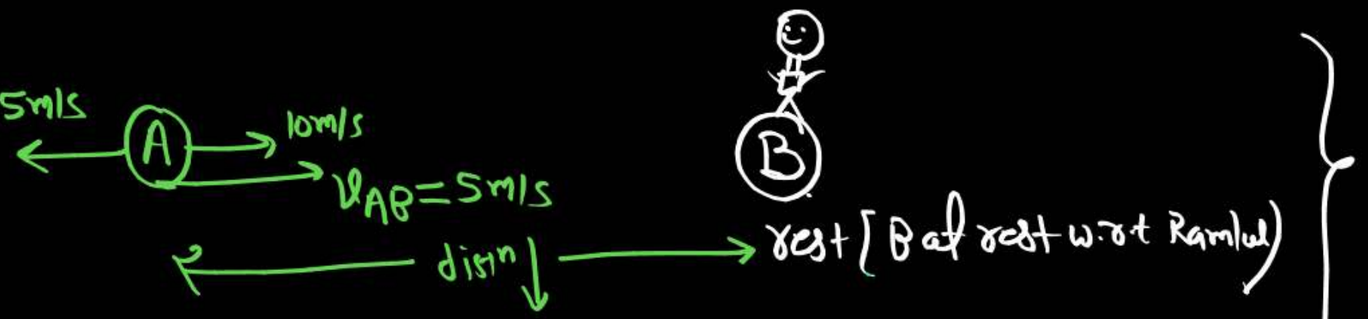
$$\vec{v}_{AB} = v_A - v_B$$
$$= 10 - 5 = 5\,m/s\,\hat{i}$$

$v_{BB} = 0$

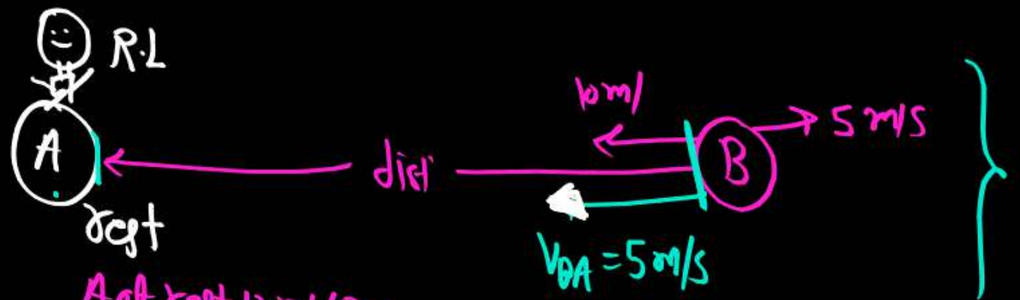
MR. BOX

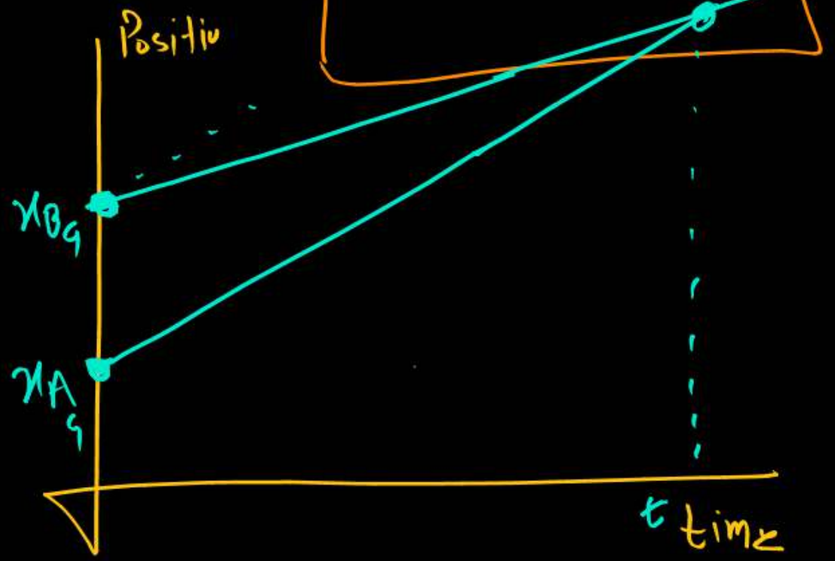
Question ko always aage wale pe built ke (observer) solve karo.

#

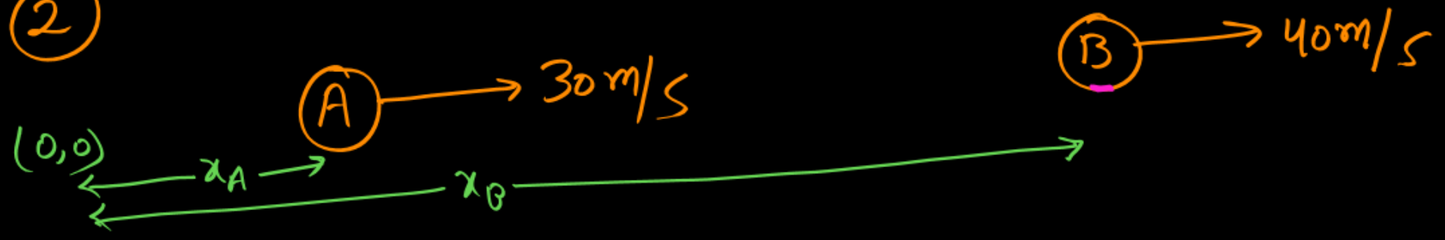


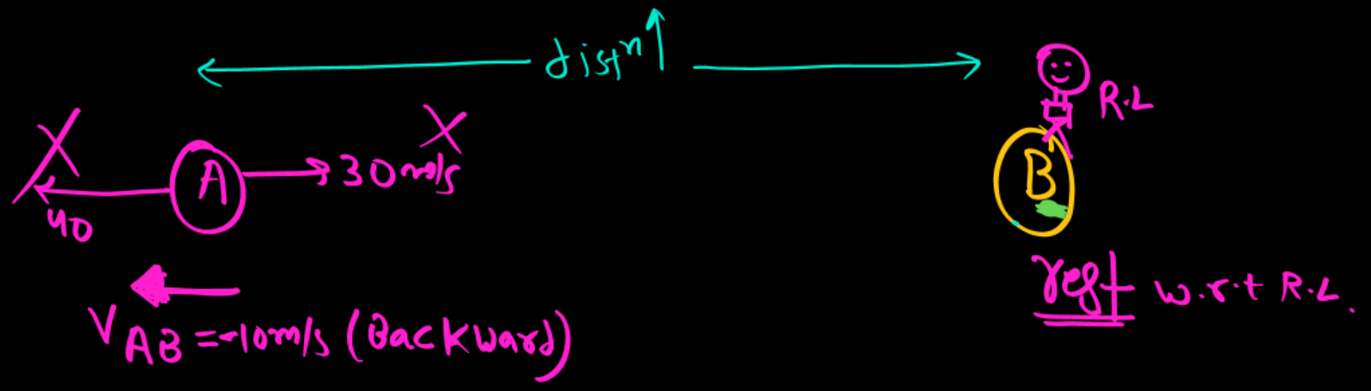
#

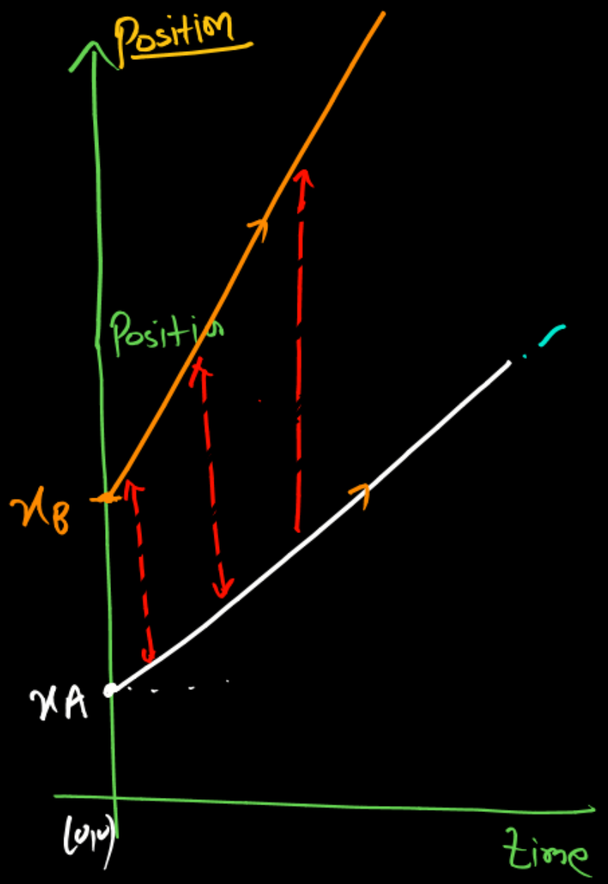




(2)


A
30 m/s
B
40 m/s
(0,0)
x_A
x_B

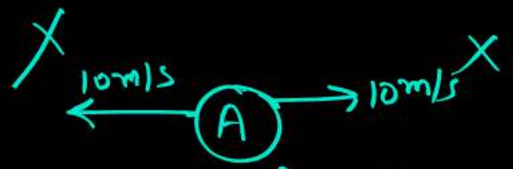

dist^n
A
30 m/s
40
R.L
B
V_{AB} = -10 m/s (Backward)
Right w.r.t R.L.


Position
Position
x_B
x_A
(0,0)
Time

3.



Relative motion

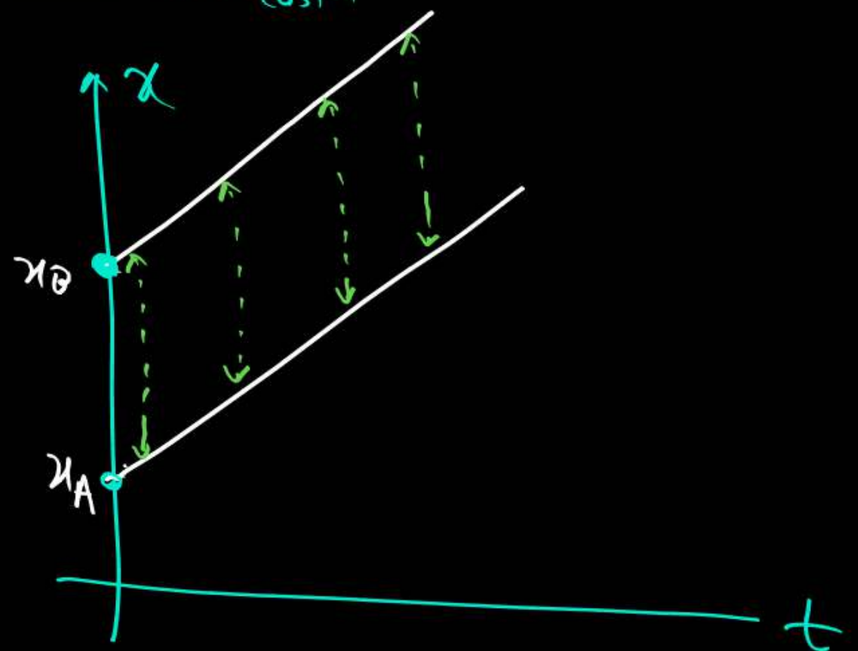


$v_{AB} = 0$

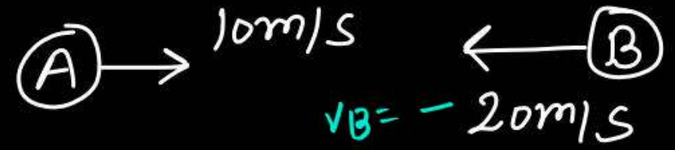


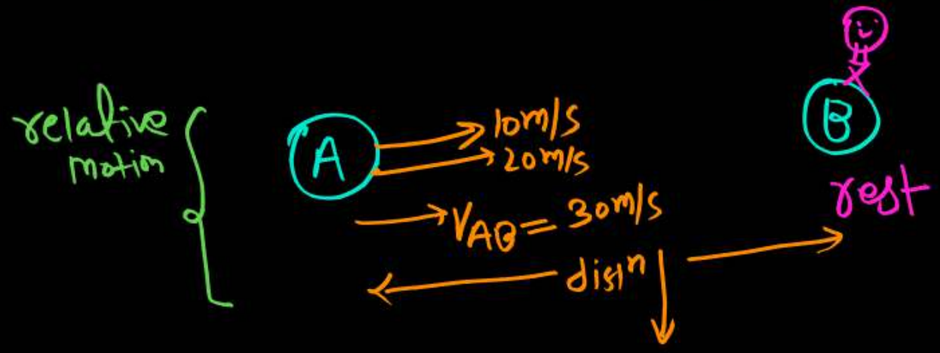


Position (x) / time graph w.r.t ground.

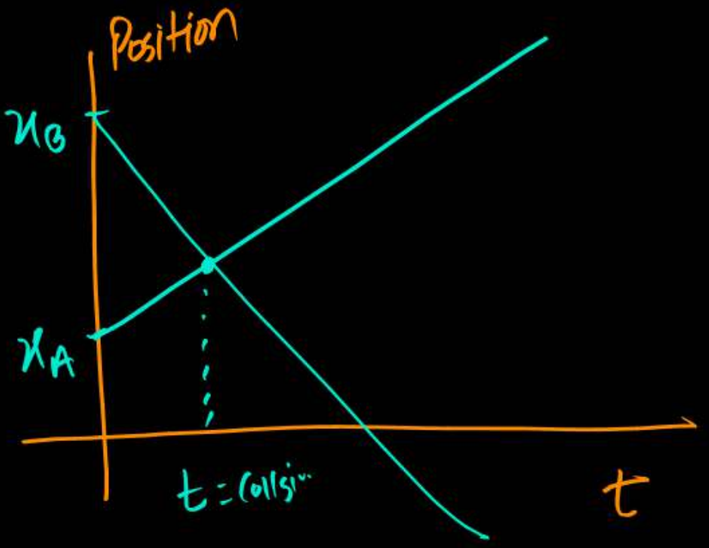


4.

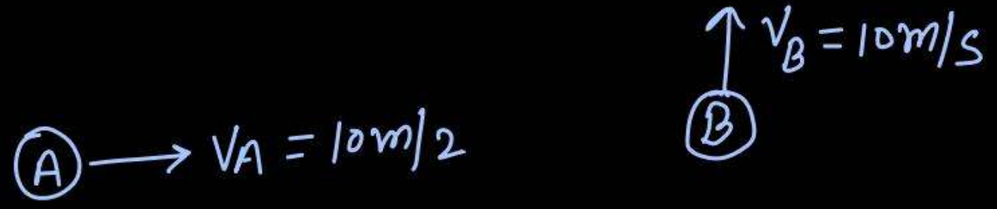




Position - time w.r.t ground

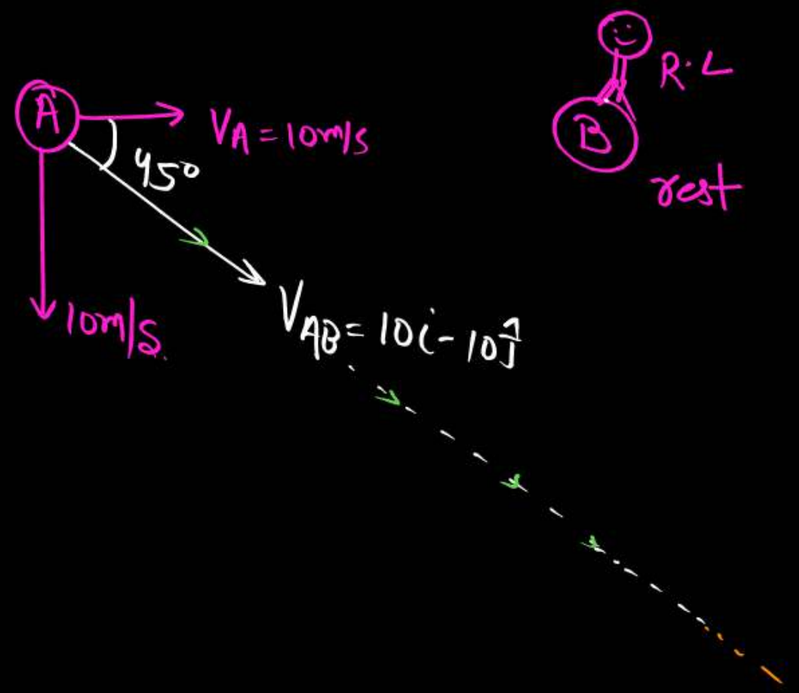


5.

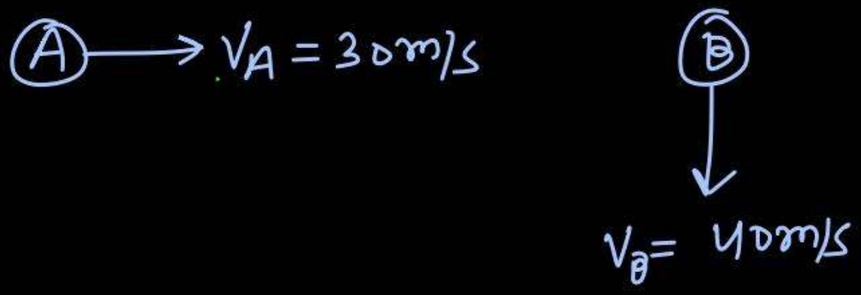


Soln

Motn of A w.r.t B

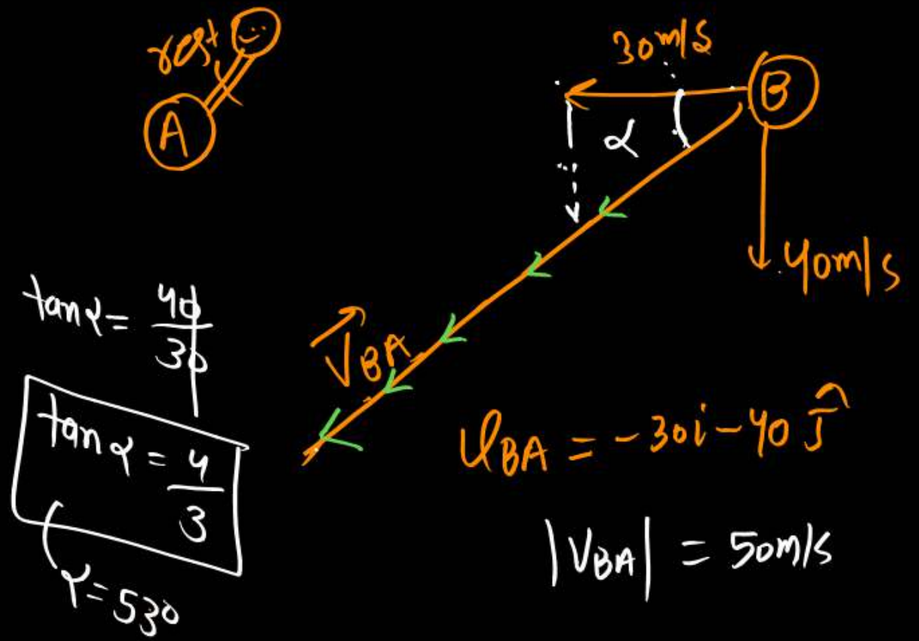


6.



Soln

Motn of B w.r.t A



$\tan\gamma = \frac{40}{30}$

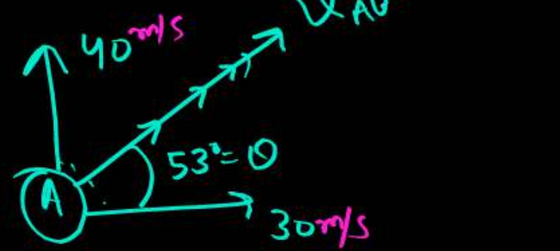
$\tan\gamma = \frac{4}{3}$

$\gamma = 53°$

(7) $V_A = 40\,m/s$ (A)

(B) $V_B = 30\,m/s$

Motion of A w.r.t (B)



(B) rest

$\tan\theta = \frac{40}{30}$

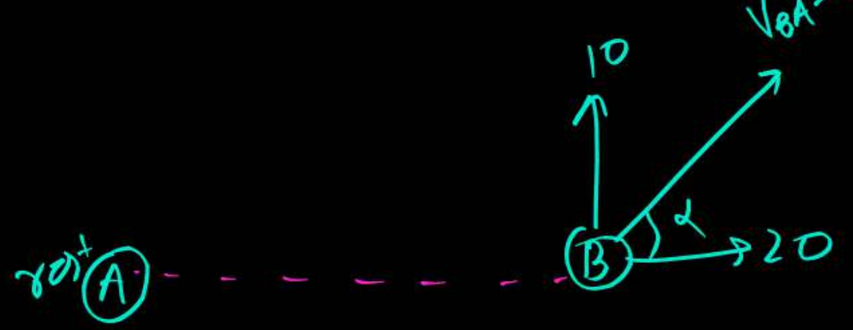
$\theta = 53°$

8.

(A) $V_A = 10\,m/s$

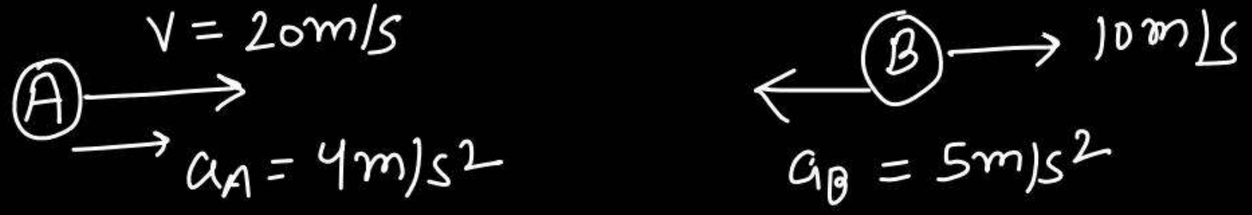
(B) $V_B = 20\,m/s$

Motion of (B) w.r.t (A)



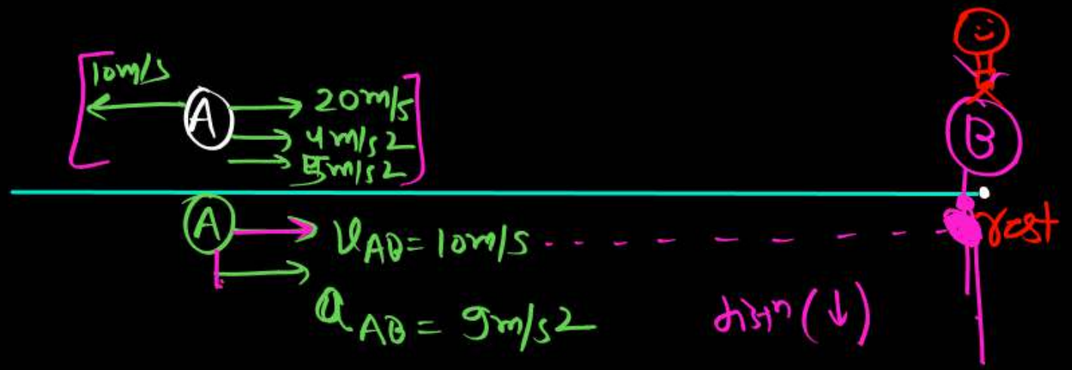
$\tan\alpha = \frac{10}{20}$

$\alpha = \tan^{-1}\left(\frac{1}{2}\right)$

9. 

Find relative velocity and retive accn. of A w.r.t. (B)

Soln

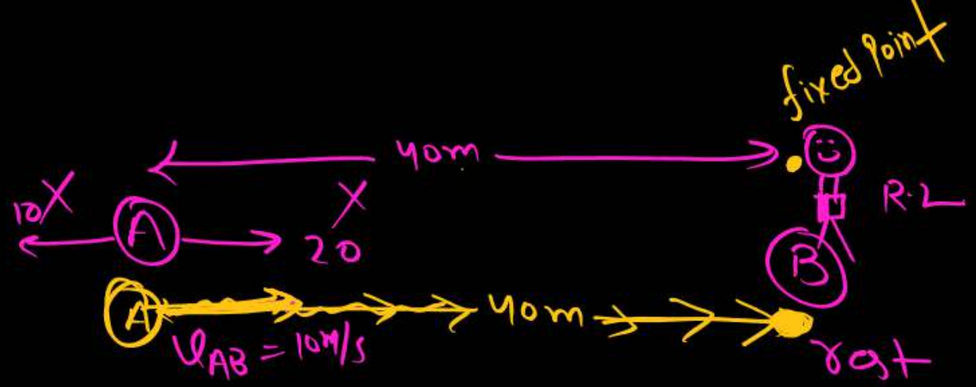


(10)



time of collision ??

Soln



$$t = \frac{40}{10} = 4sec$$

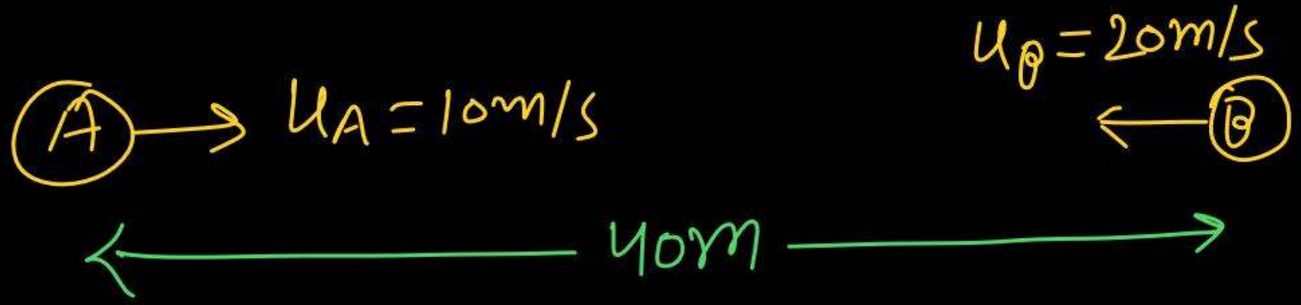
(#) dist^n moved by A w.r.t ground at time of collision from (t=0)

$x_A = v_A t = 20 \times 4 = 80m.$

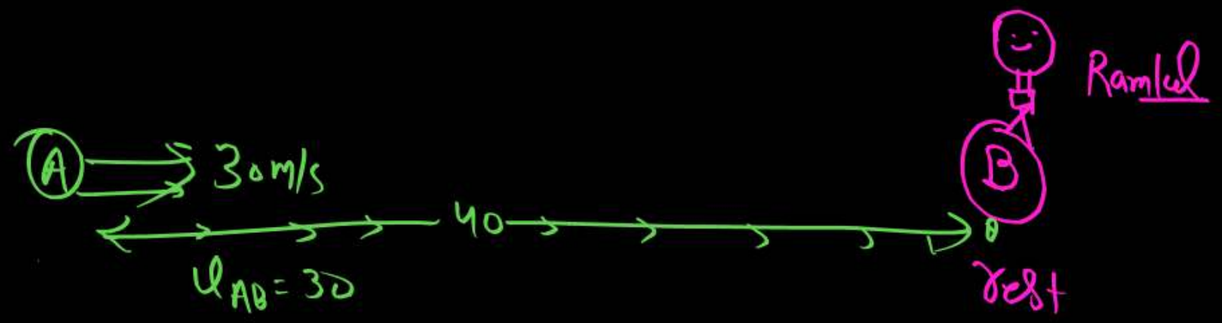
$x_B = 10 \times 4 = 40m.$

(11)

Time of collision.



Sol^n

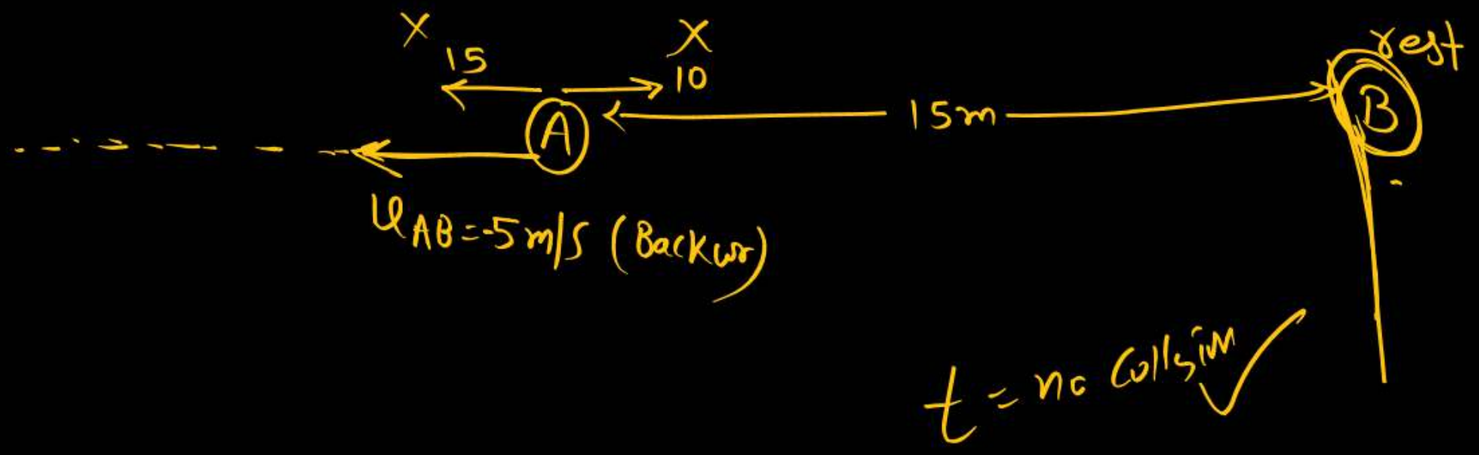


$$t = \frac{40}{30} = \frac{4}{3} \text{ sec.}$$

(12)



time when they meet.



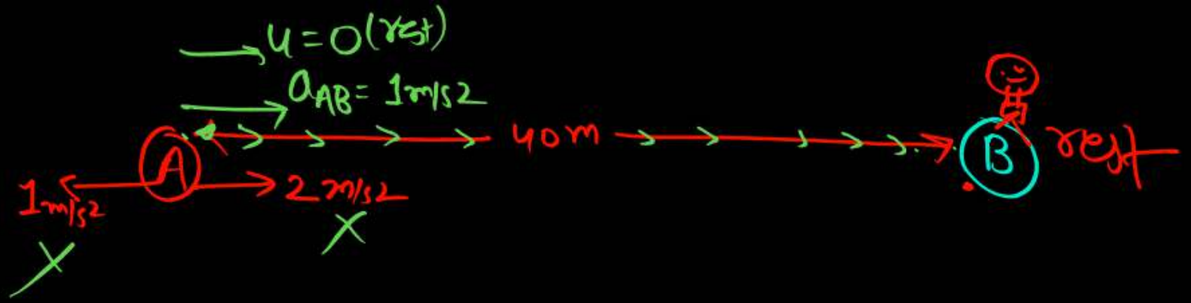
t = no collision ✓

(13)

$t=0$



Time when they meet

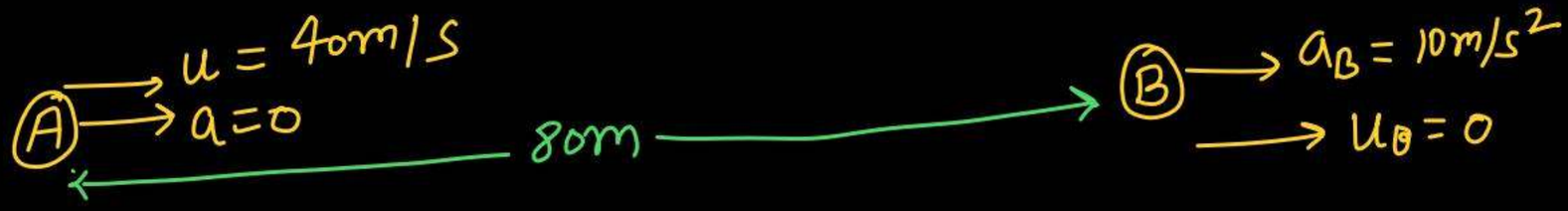


$$S_{AB} = u_{AB} t + \frac{1}{2} a_{AB} t^2$$

$$40 = 0 + \frac{1}{2} 1 (t)^2$$

$t = \sqrt{80}$ sec

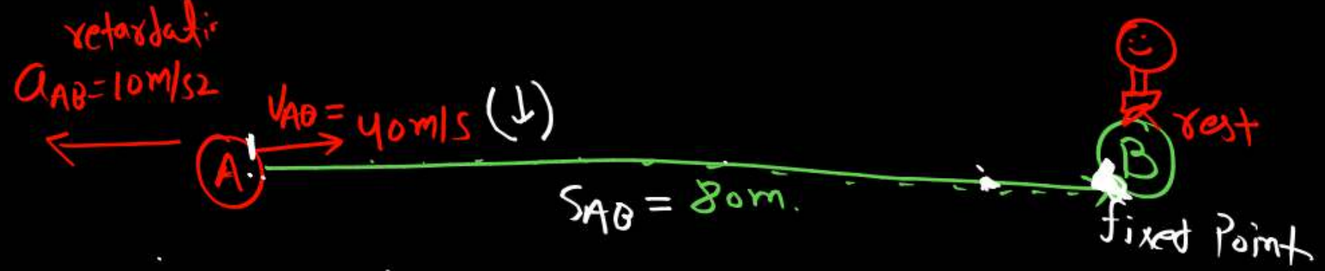
(14) $t=0$



time when they meet.

(a) 4 sec
(b) 2 sec
(c) 16 sec
(d) None of these

Soln



$$80 = 40t - \frac{1}{2}(10)t^2$$

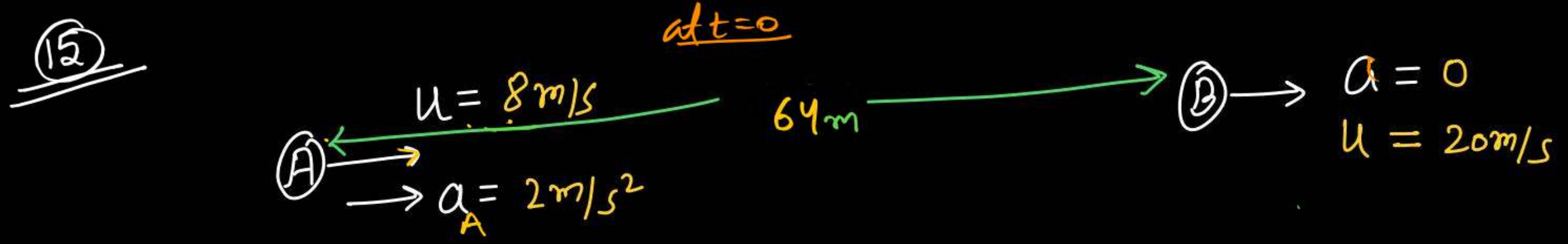
$$5t^2 - 40t + 80 = 0$$

$$t^2 - 8t + 16 = 0$$

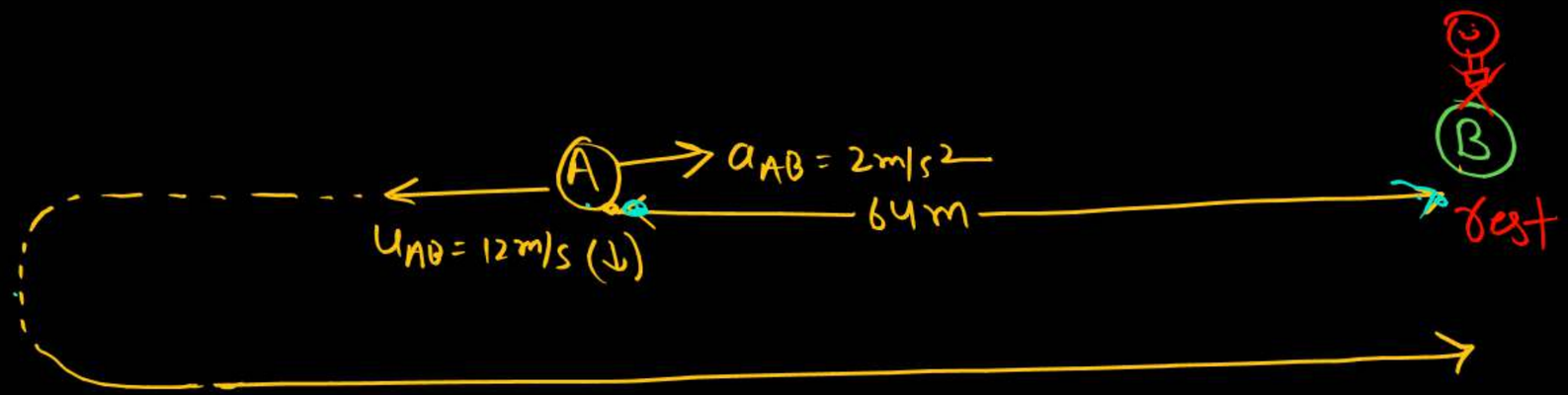
$$t^2 - 4t - 4t + 16 = 0$$

$t = 4$ sec

(12)



Soln



$s = ut + \frac{1}{2}at^2$ ⟵ for A w.r.t B

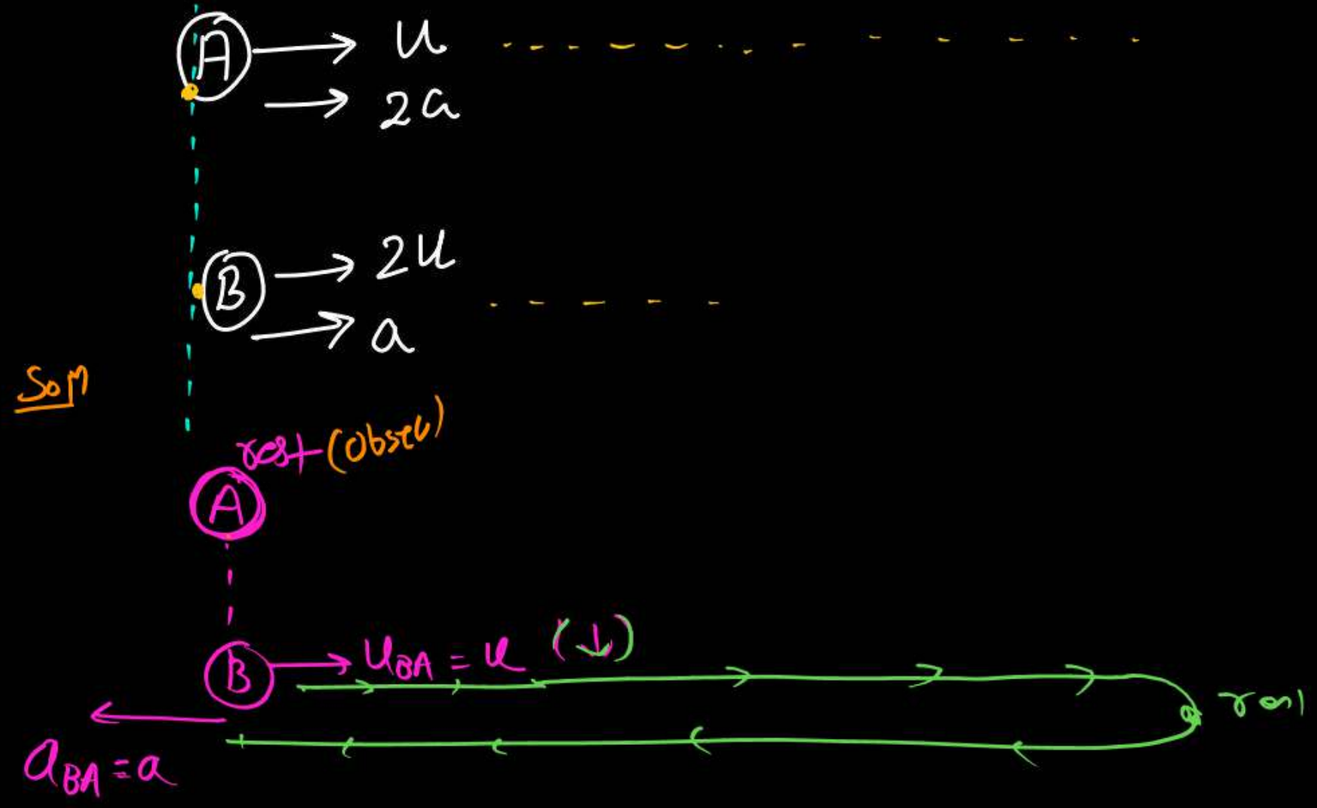
$64 = -12t + \frac{1}{2}\times 2\, t^2$

$t^2 - 12t - 64 = 0$

$t^2 - 16t + 4t - 64 = 0$

$t_1 = 16$ sec ✓

(16) At $t=0$ what time they meet again.



(a) never meet (−1 pen)

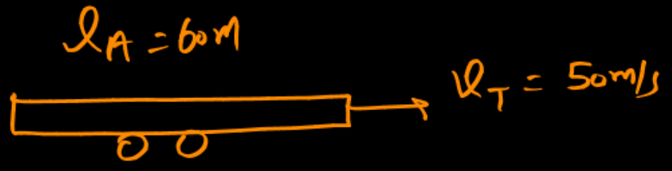
(b) will meet (+1 pen)

$$s = ut + \frac{1}{2}at^2$$

$$0 = ut - \frac{1}{2}at^2$$

$$t = \frac{2u}{a} \qquad \frac{1}{2}at^2 = ut$$

17



$$t = \frac{l_{Train}}{v_T}$$

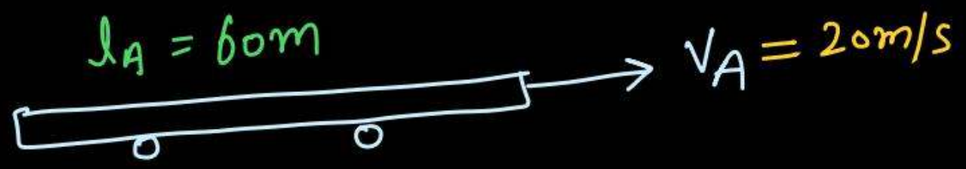
$$= \frac{6\cancel{0}}{5\cancel{0}}$$

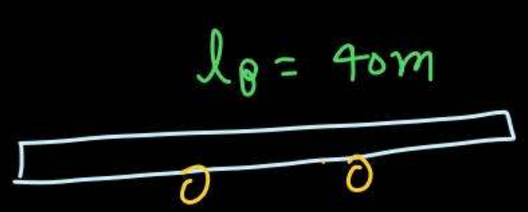
$$= \frac{6}{5} \text{ sec}$$



$$t_{(rossiy} =$$

(18)



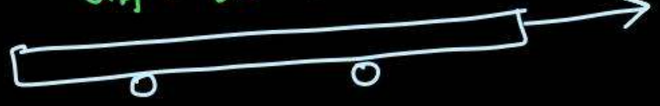


$t_{cross} = ?$

$= \frac{l_A + l_B}{20}$

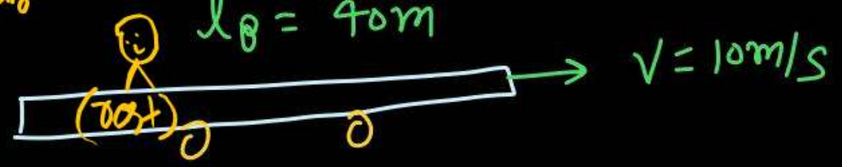
$= \frac{100}{20} = 5$

(19)

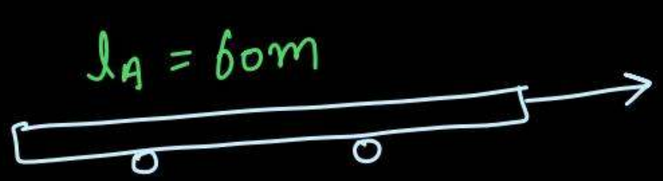


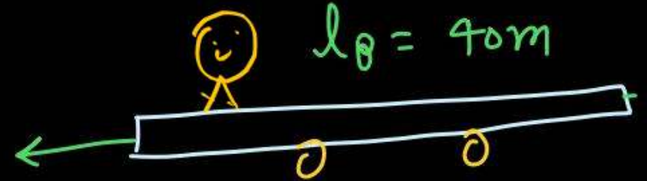
$t = \left(\frac{60+40}{20-10}\right) = \frac{100}{10} = 10sec$

(overtake)



(20)

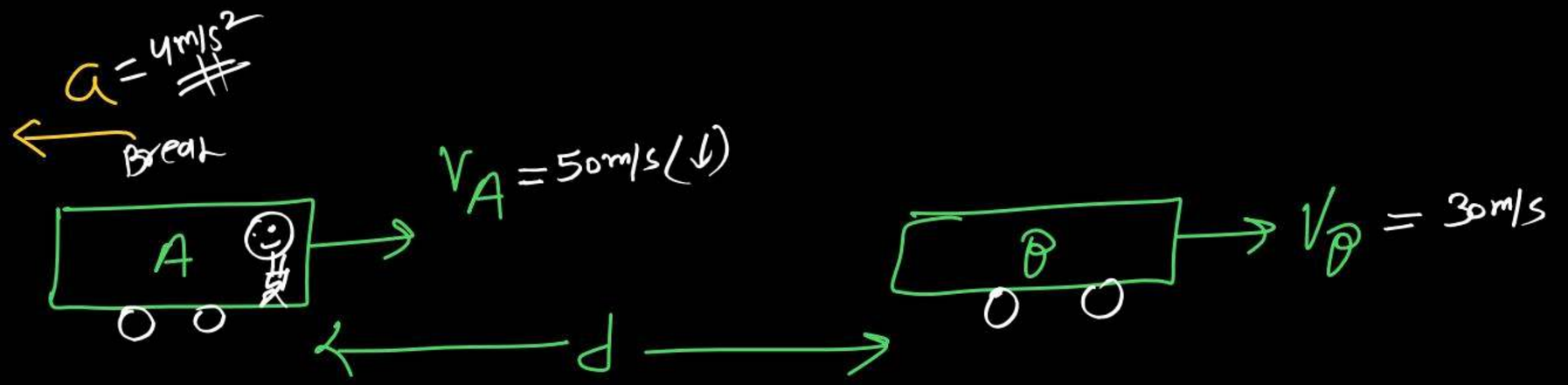




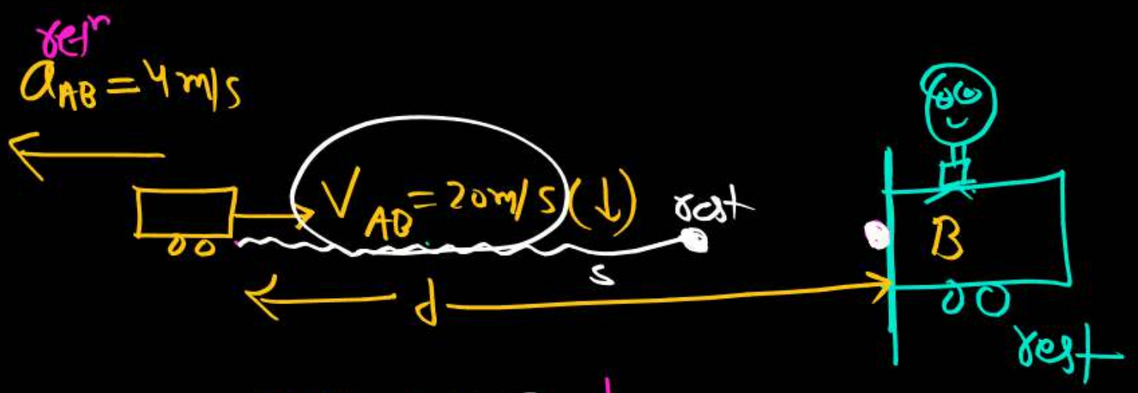
$t = \frac{60+40}{20+10}$

(Ans) $= \frac{100}{30} = \frac{10}{3} = 3.3sec$

(21)



Condition for No Collision :-



$S(\text{toppis}) \leq d$

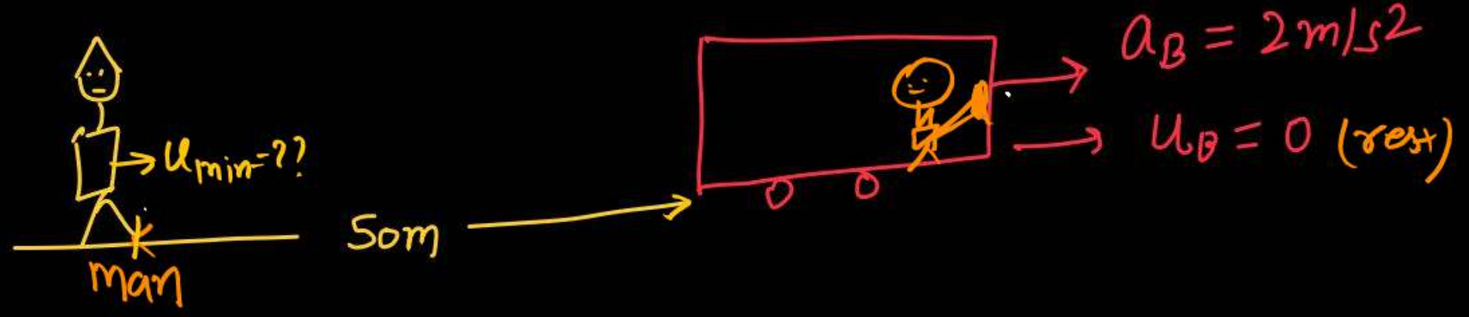
$\frac{u_{AB}^2}{2(a_{AB})} \leq d$

So

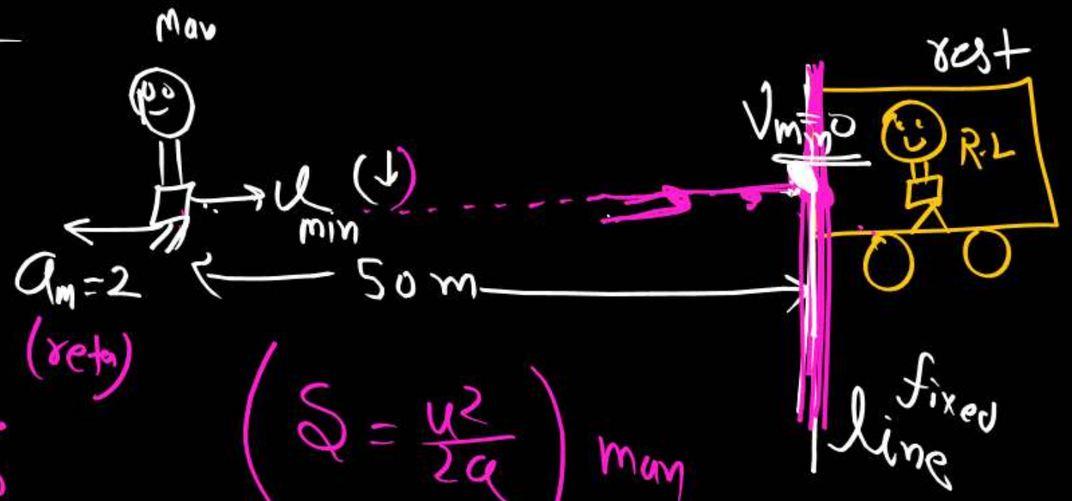
$\frac{400}{2 \times 4} \leq d$

$50 \leq d$

(22) find minimum velocity of man so that he can catch the bus which is moving with rest and acceleration $2\ m/s^2$ & 50m ahed of man.



Soln



$$\left(S = \frac{u^2}{2a}\right)_{man}$$

$$50 = \frac{u^2}{2\times 2}$$

$$u = \sqrt{200}$$

$$u = 10\sqrt{2}\ m/s$$

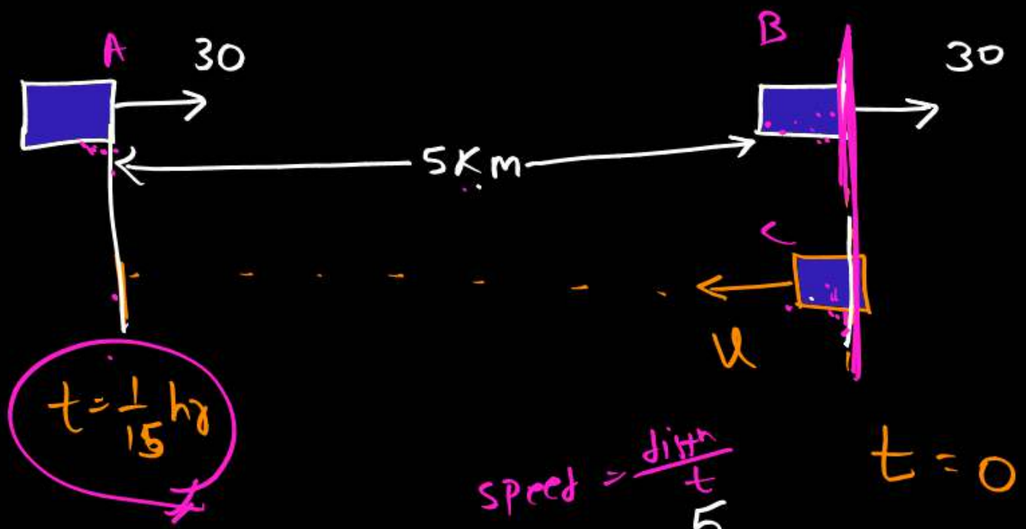
Question

23 (Copy)

Two cars are moving in the same direction with a speed of 30 km/h. They are separated from each other by 5 km. Third car moving in the opposite direction meets the two cars after an interval of 4 minutes. The speed of the third car is

$t = 4\text{min} = \frac{4}{60} = \frac{1}{15}\text{hr}$

1. 30 km/h
2. 25 km/h
3. 40 km/h
4. 45 km/h



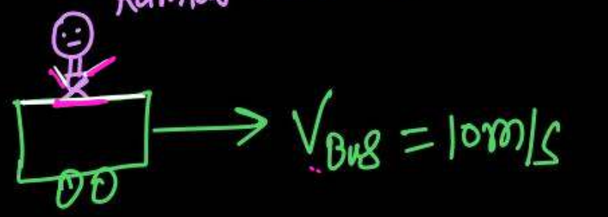
$\text{speed} = \frac{\text{dist}^n}{t}$

$30 + u = \frac{5}{1/15}$

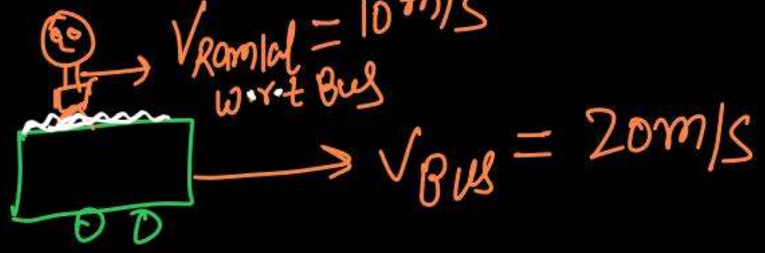
$30 + u = 75$

$u = 75 - 30 = 45\ \text{km/hr}$

Relative Motion from moving frame :-

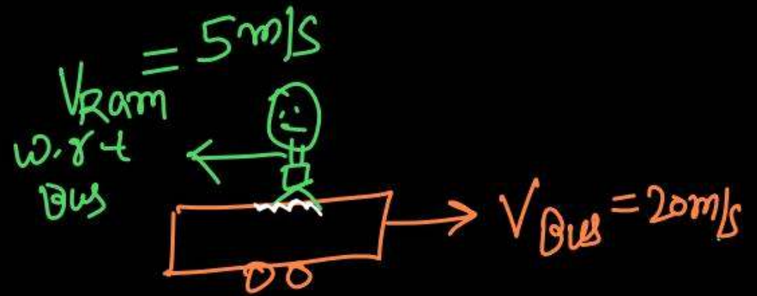


$$\left[\begin{array}{l} v_{Ramlal\ ground} = 10 m/s \\ v_{Ramlal\ w.r.t\ Bus} = v_{R_g} - v_{B_g} \\ \qquad = 10 - 10 \\ \qquad = 0 \end{array}\right]$$



$$v_{Ramlal/Bus} = 10 m/s$$

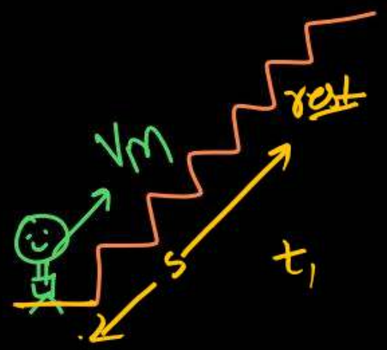
$$v_{Ramlal/ground} = 10 + 20 = 30 m/s$$



$$v_{Ram\ w.r.t\ (Bus)} = -5\,i$$

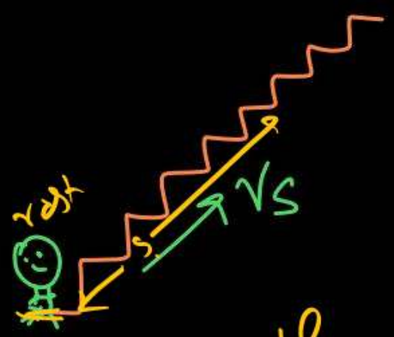
$$v_{Ram\ ground} = -5 + 20 = 15 m/s\,i$$

escalator at rest



$$t_1 = \frac{s}{v_m} \quad -①$$

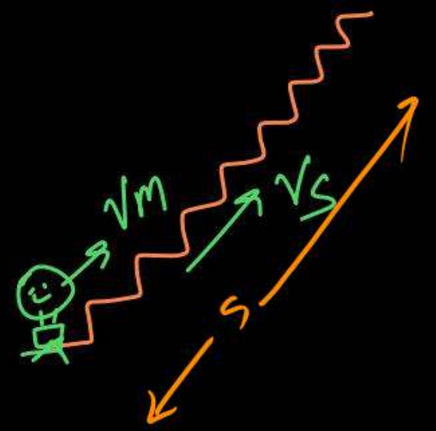
escalator moving man at rest



$v_{man\ gain} = v_s$

$$t_2 = \frac{s}{v_s}$$

escalator & man both are moving



$$t_3 = \frac{s}{v_m + v_s}$$ ✓

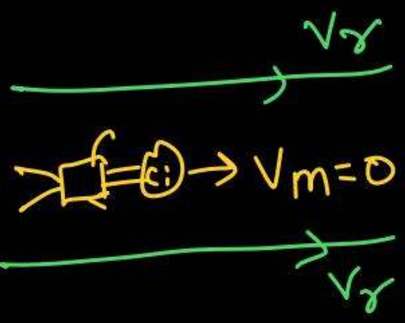
Reln B/w t_1 & t_2 & t_3

Question

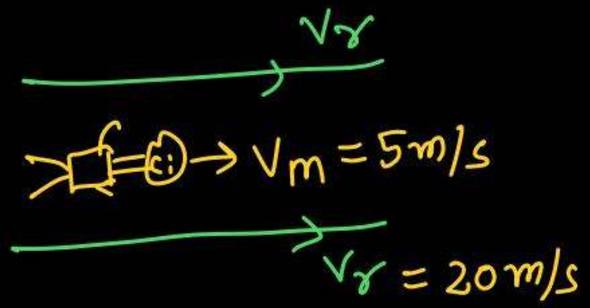
Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be **[NEET -2017]**

1. $\frac{t_1 + t_2}{2}$
2. $\frac{t_1 t_2}{t_2 - t_1}$
3. $\frac{t_1 t_2}{t_2 + t_1}$
4. $t_1 - t_2$

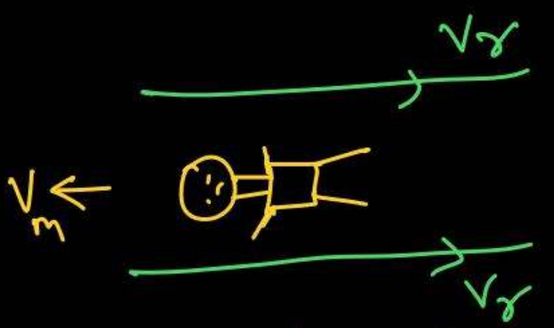
River man in 1-D



man is not swiming



man is swiming with v_m



Man is swiming backward.

Question

Two trains each of length 100 m moving parallel towards each other at speed 72 km/h and 36 km/h respectively. In how much time will they cross each other?

1. 4.5 s
2. 6.67 s
3. 3.5 s
4. 7.25 s

Question

A bus is moving with a speed of 10 ms^{-1} on a straight road. A scooterist wishes to overtake the bus in 100 s. If the bus is at a distance of 1 km from the scooterist, with what speed should the scooterist chase the bus? **[2009]**

1. 40 ms^{-1}
2. 25 ms^{-1}
3. 10 ms^{-1}
4. 20 ms^{-1}

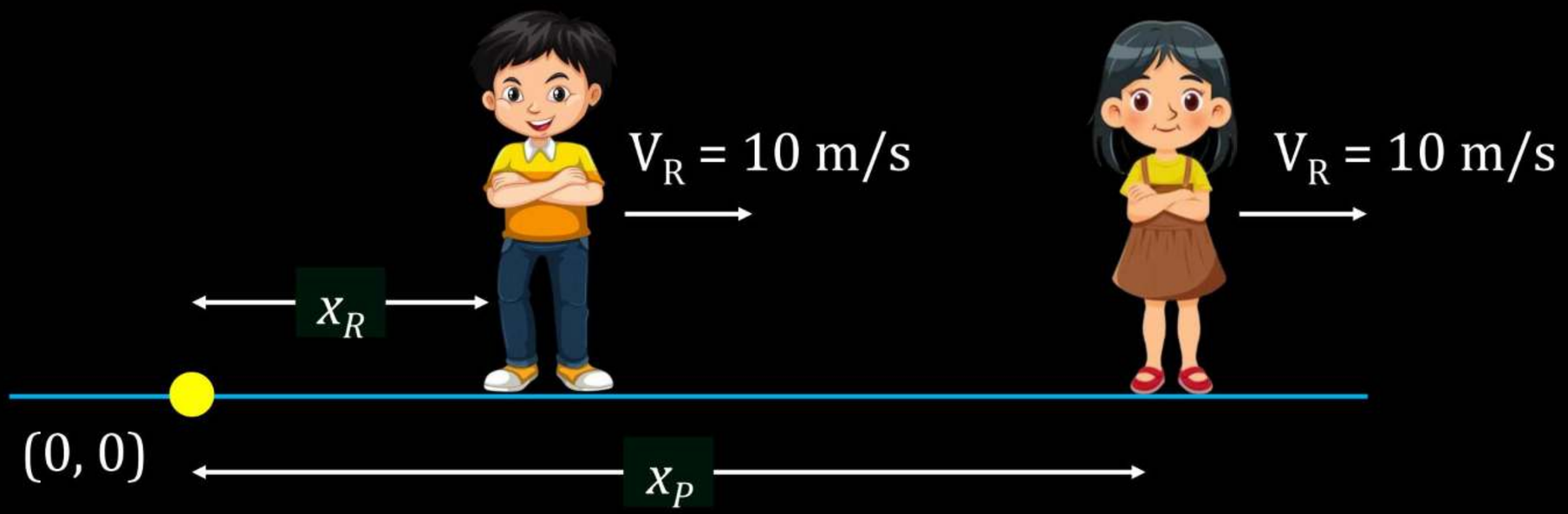
Question

Two cars A and B are moving in same direction with velocities 30 m/s and 20 m/s. When car A is at a distance d behind the car B, the driver of the car A applies brakes producing uniform retardation of 2 m/s^2. There will be no collision when

1. $d < 2.5$ m
2. $d > 125$ m
3. $d > 25$ m
4. $d < 125$ m

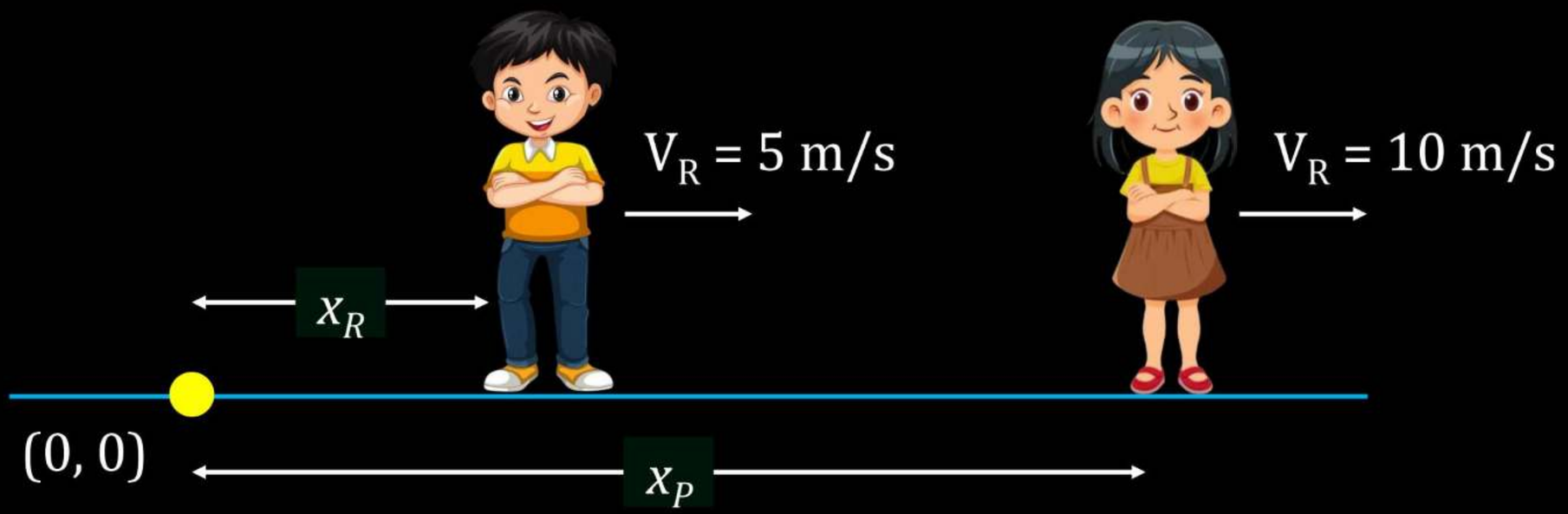
Question

Draw position-time graph and comment on relative separation?



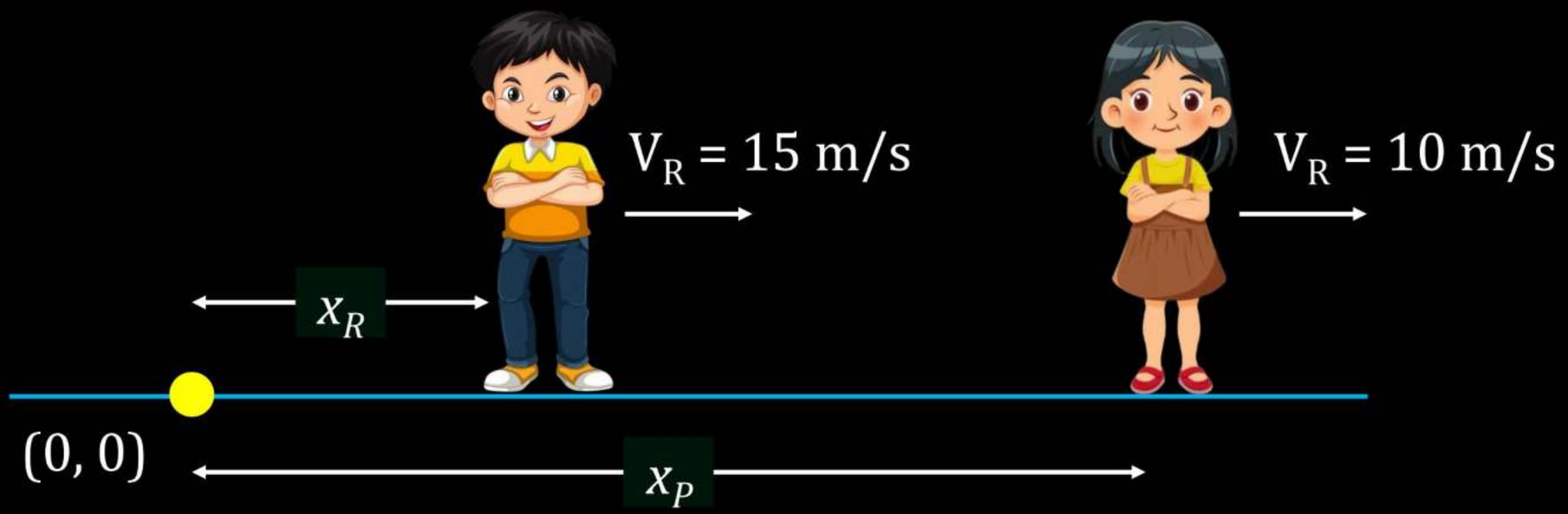
Question

Draw position-time graph and comment on relative separation?



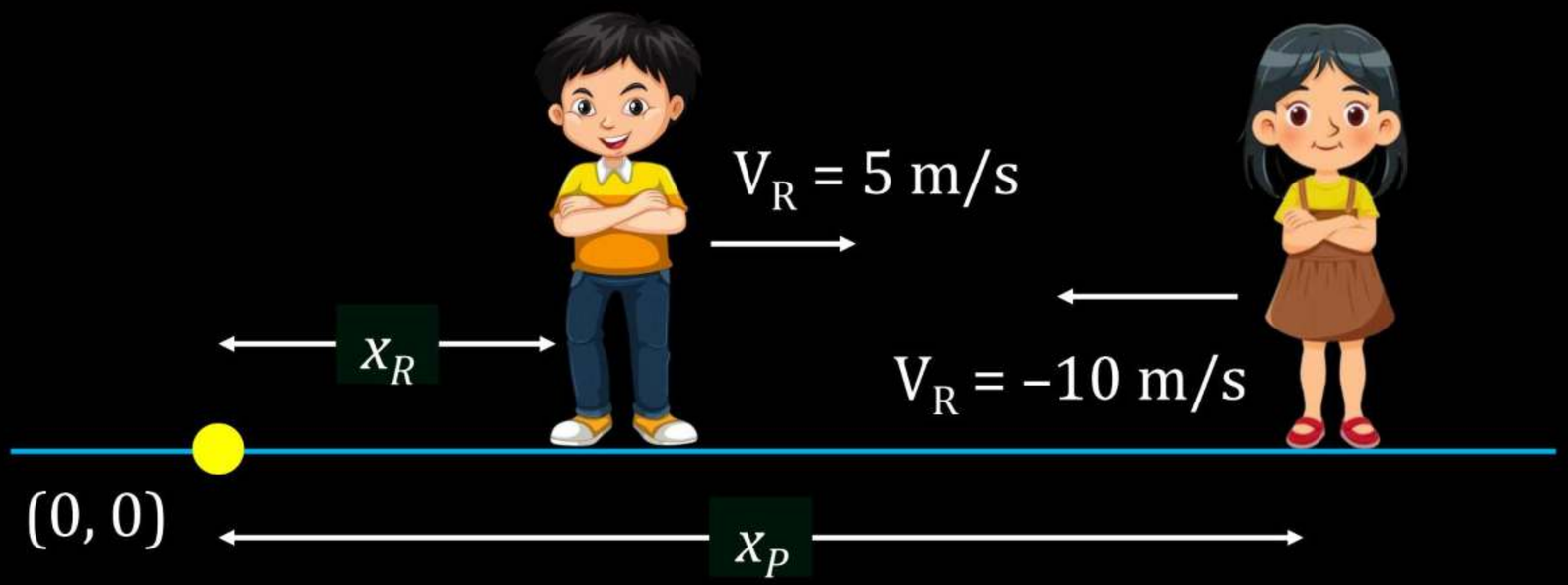
Question

Draw position-time graph and comment on relative separation?



Question

Draw position-time graph and comment on relative separation?



Question

Bus is moving with constant velocity 21 m/s and Ramlal starts his motion from rest and constant acceleration 2 m/s^2. If initial distance is 100 m then find time when Ramlal will catch the bus.

Question

Just Try

Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses. **[NEET 2025]**

1. 9 min, 40 km/h
2. 25 min, 100 km/h
3. 10 min, 90 km/h
4. 15 min, 120 km/h

Thank

You